2. In the use-dilution test, a series of tubes contain increasing concentrations of disinfectant; the more the chemical can be diluted and still be effective, the higher its rating.
3. In the filter paper method, a disk of filter paper is soaked with a chemical and placed on an inoculated agar plate; a clear zone of inhibition indicates effectiveness.

Agar Plate Method

A plate of agar medium is inoculated with the test organism and the chemical agent is placed on the surface of the medium. The chemical solution is first impregnated in absorbent papers or confined by a hollow cylinder placed on the agar surface. Following incubation, the plate is observed for a zone of inhibition around the chemical agent. This is particularly suiTable for semisolid preparations.

Tube Dilution Methods

Appropriately diluted water soluble liquid substances are dispensed into sterile test tubes and are inoculated with a measured amount of the test organism. At specified intervals, a transfer is made from this tube into tubes of sterile media that are then incubated and observed for the appearance of growth. It is necessary in this type of procedure to ascertain whether the inhibitory action is bactericidal and not bacteriostatic. This approach can also be used to determine the number of organisms killed per unit time by performing a plate count on samples taken at appropriate intervals.

Phenol Coefficient Test

Phenol coefficient is a measure of the bactericidal activity of a chemical compound in relation to phenol. Phenol coefficient is calculated by dividing the concentration of test disinfectant at which it kills the organism by the concentration of phenol at which it kills the organism in 10 minutes and not in 5 minutes under the same conditions. This method is used for evaluating the efficiency of water-miscible disinfectants.

Series of 10 test tubes with 2ml of distilled water is taken (Figure 3.1). Phenol is added to first test tube and dilution is made by transferring 1ml to next tube up to 5 dilutions. Similarly commercial disinfectant is also diluted. Pure culture of test organisms, such as *Staphylococcus aureus* or *Salmonella typhi*, is added to test tubes. Subcultures from these tubes incubated at 37°C for 48 hours are examined for the presence or absence of growth at intervals of 5, 10 and 15 minutes. The highest dilution that kills the bacteria after 10 minutes, but not after 5 minutes is used to calculate the phenol coefficient (Table 3.3),

Table 3.3: Illustration of phenol coefficient determination

Chemical Agent and Dilution	Presence of Growth in Subcultures (minutes)		
	5	10	15
Phenol			
1:80	-	-	-
1:90*	+	-	-
1:100	+	+	-
Test Chemical			
1:400	-	-	-
1:450+	+	-	-
1:500	+	+	-

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